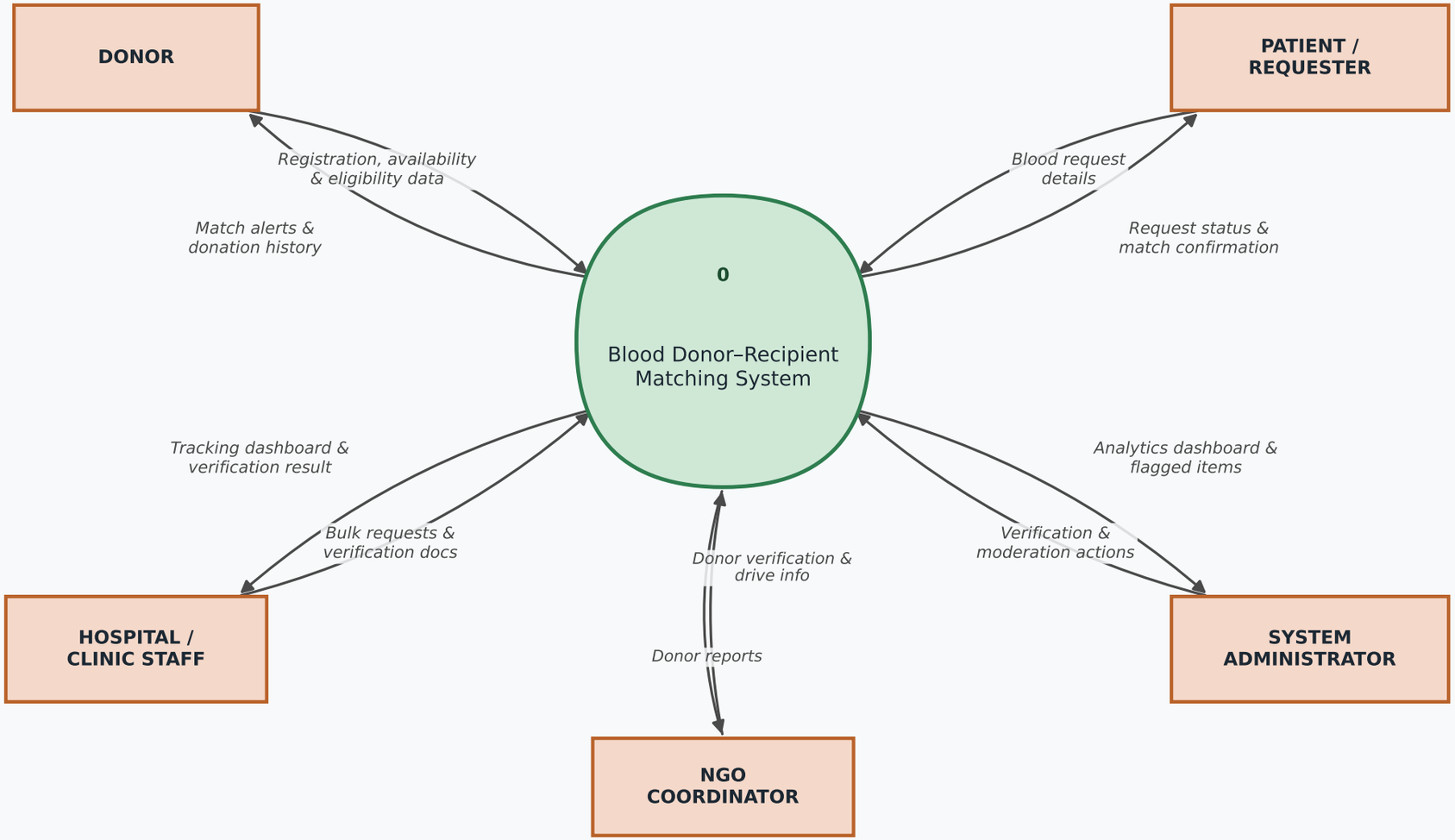
Blood Donor–Recipient Matching System

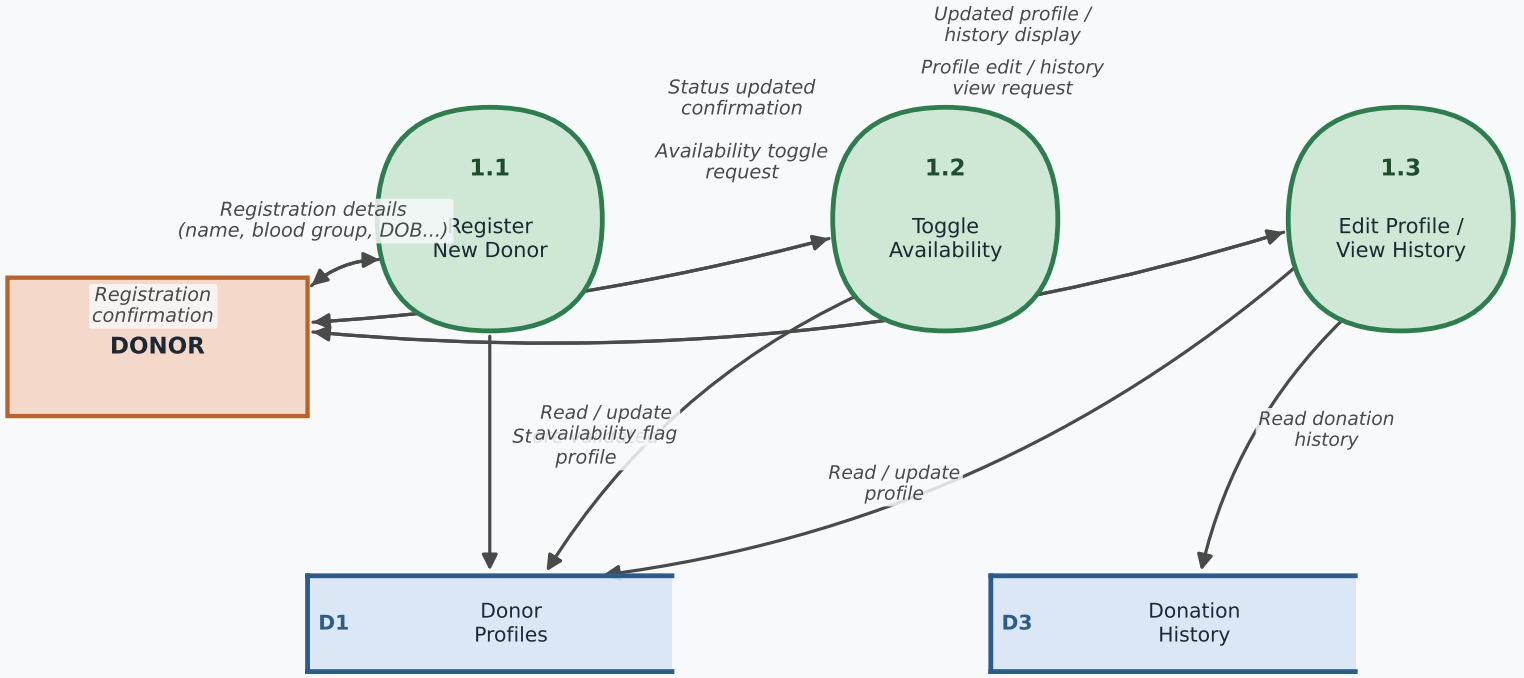
Level 0 DFD — Context Diagram



Legend: rectangle = external entity | rounded shape = process | open box = data store | arrows = data flows

1.0 Donor Registration and Profile Management

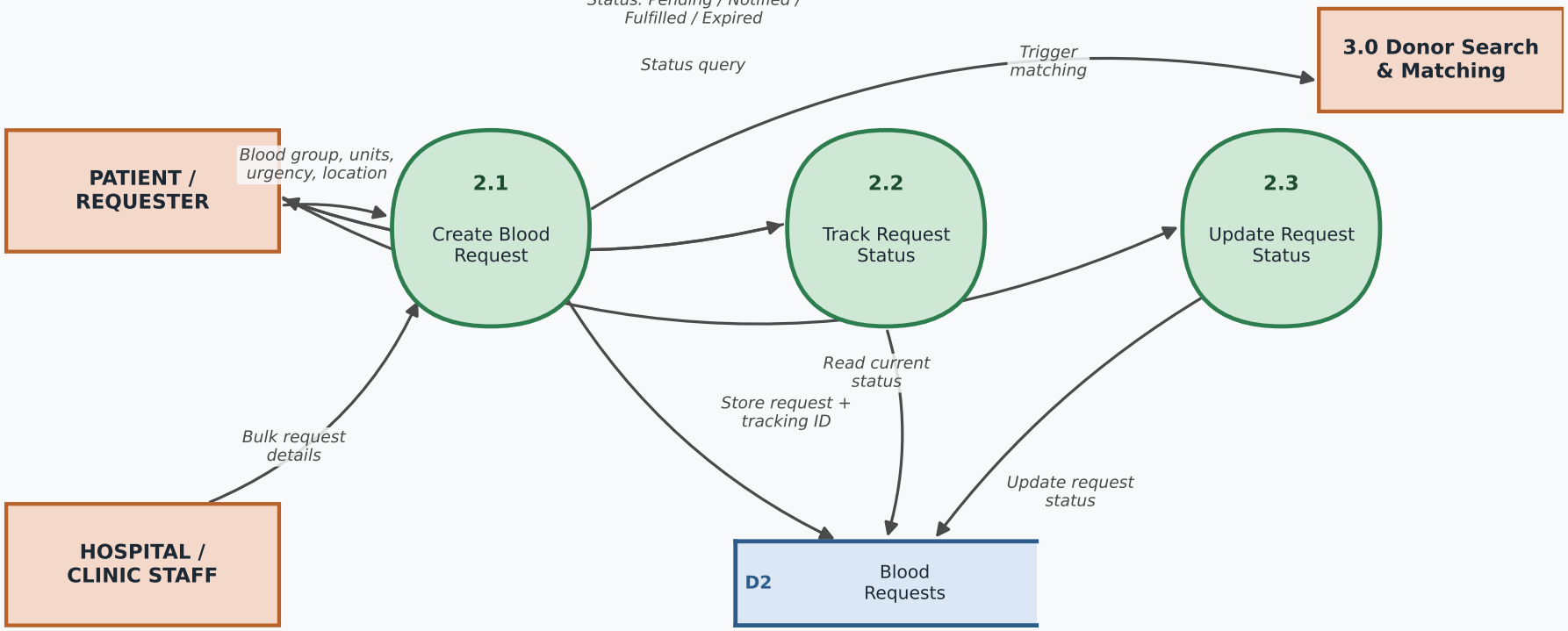
Level 1 DFD — decomposition of Process 1 (REQ-4.1.1 – 4.1.5)



Legend: rectangle = external entity | rounded shape = process | open box = data store | arrows = data flows

2.0 Blood Request Posting

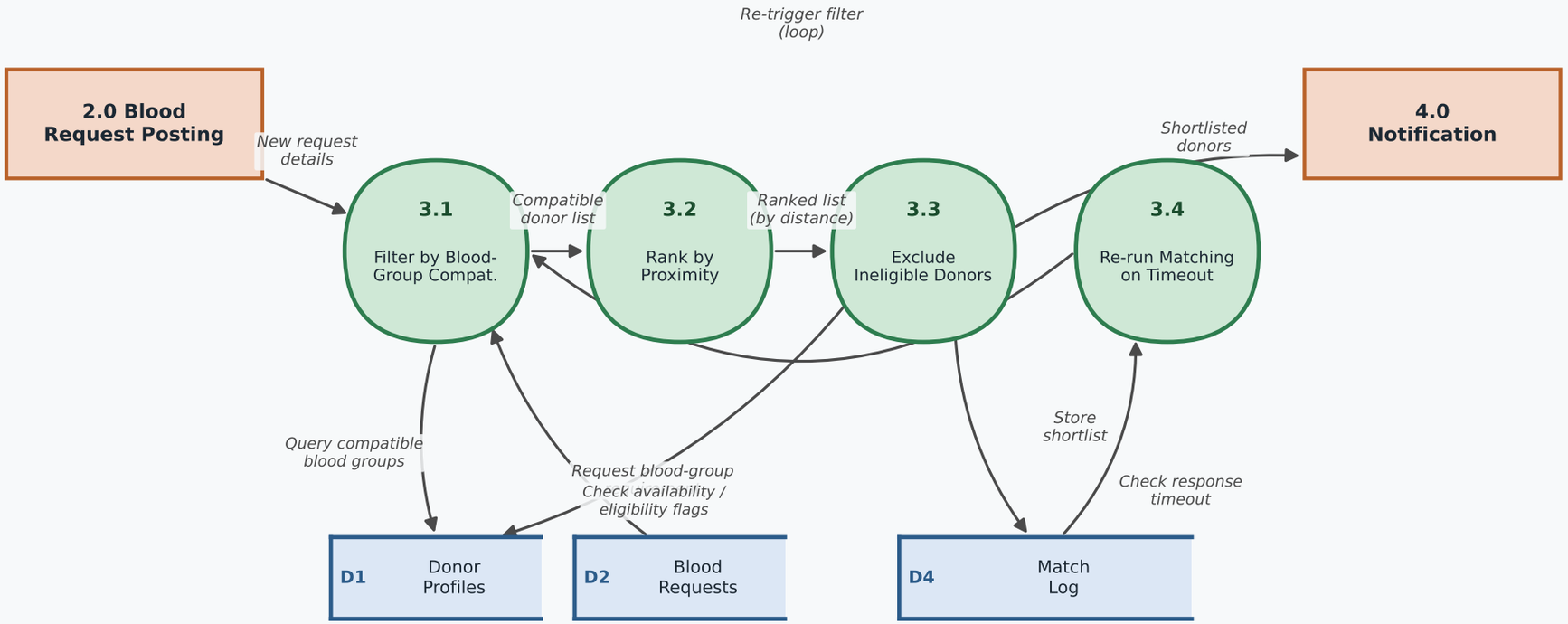
Level 1 DFD — decomposition of Process 2 (4.2.1 – 4.2.4)



Legend: rectangle = external entity | rounded shape = process | open box = data store | arrows = data flows | dashed-look box '3.0' = reference to another Level-1 process

3.0 Donor Search and Matching

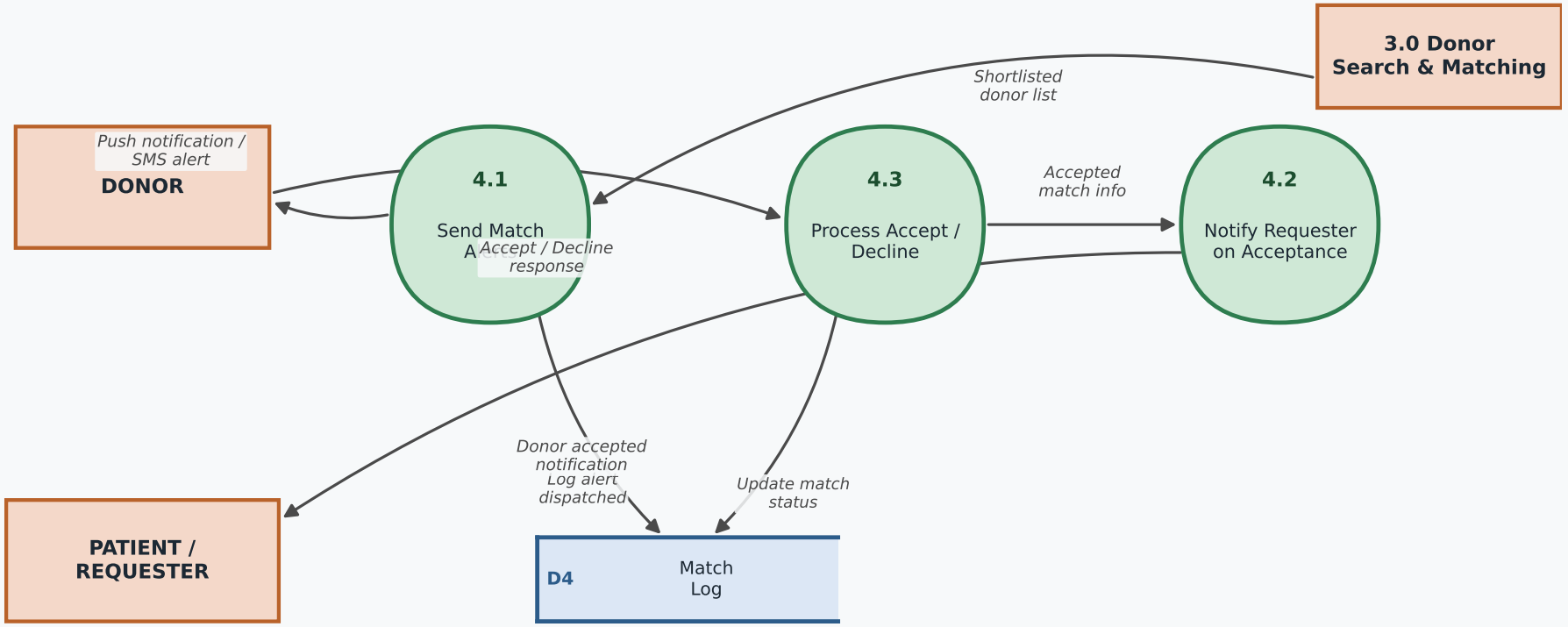
Level 1 DFD — decomposition of Process 3 (REQ-4.3.1 – 4.3.4)



Legend: rectangle = external entity | rounded shape = process | open box = data store | arrows = data flows | side boxes '2.0'/'4.0' = references to other Level-1 processes

4.0 Notification System

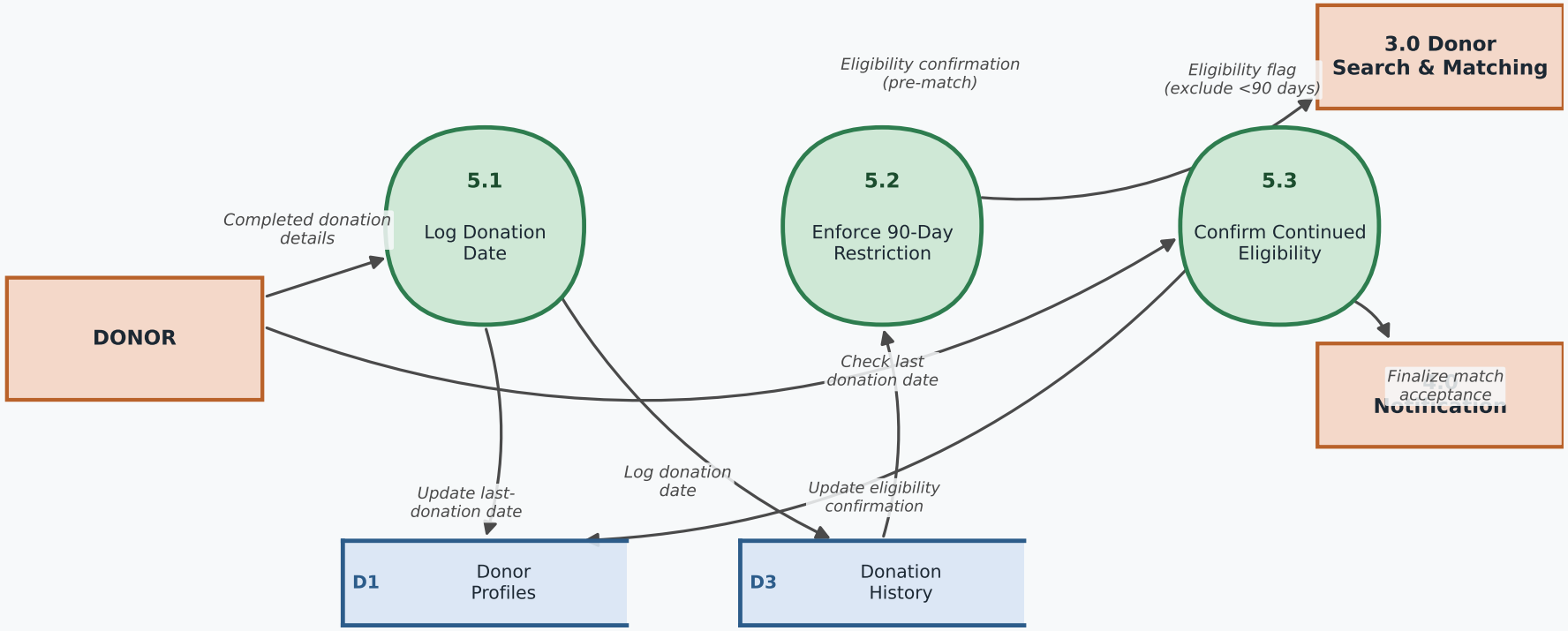
Level 1 DFD — decomposition of Process 4 (REQ-4.4.1 – 4.4.3)



Legend: rectangle = external entity | rounded shape = process | open box = data store | arrows = data flows | side box '3.0' = reference to another Level-1 process

5.0 Donor Eligibility Verification

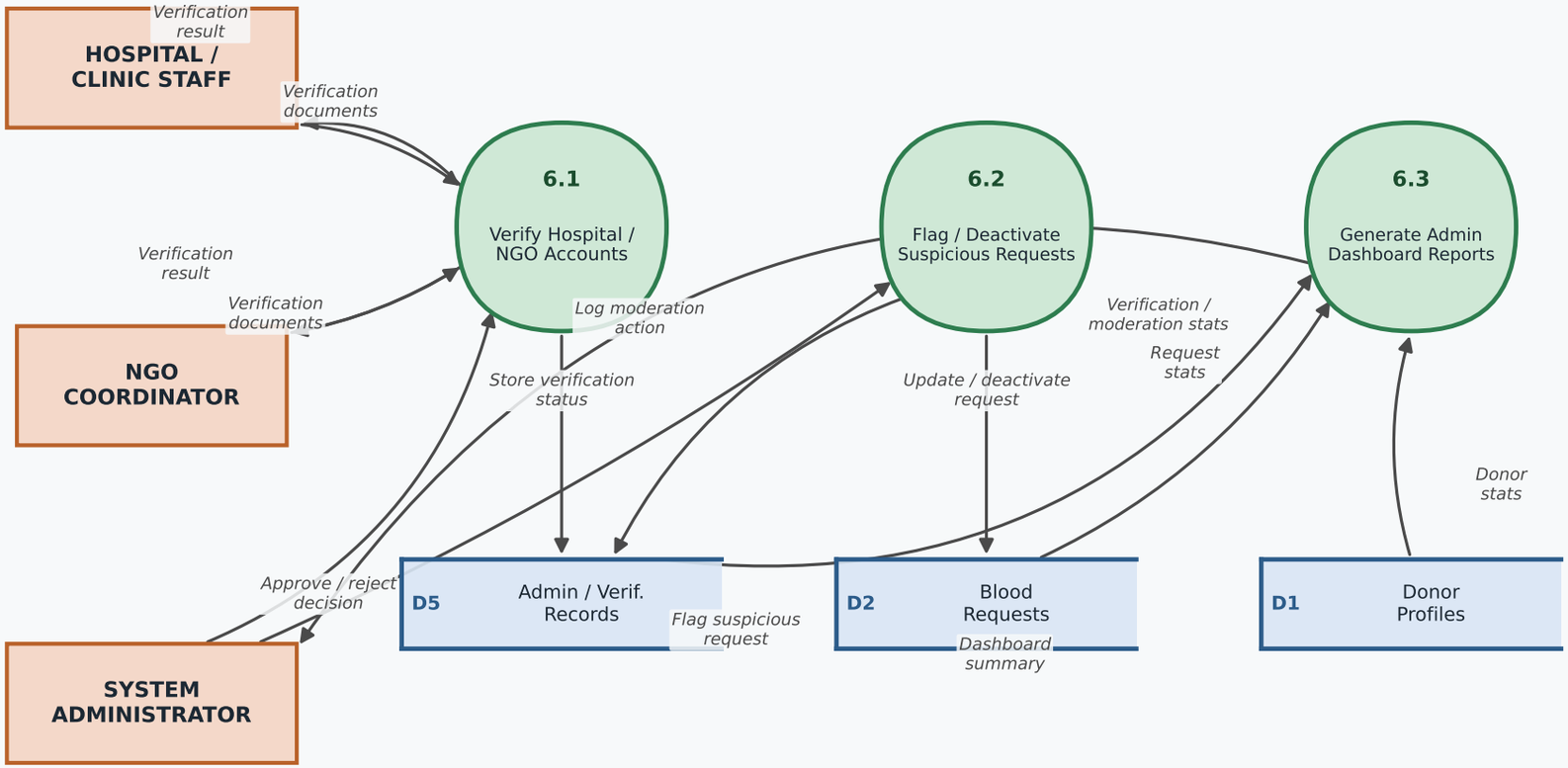
Level 1 DFD — decomposition of Process 5 (REQ-4.5.1 – 4.5.3)



Legend: rectangle = external entity | rounded shape = process | open box = data store | arrows = data flows | side boxes '3.0'/'4.0' = references to other Level-1 processes

6.0 Admin Verification and Moderation

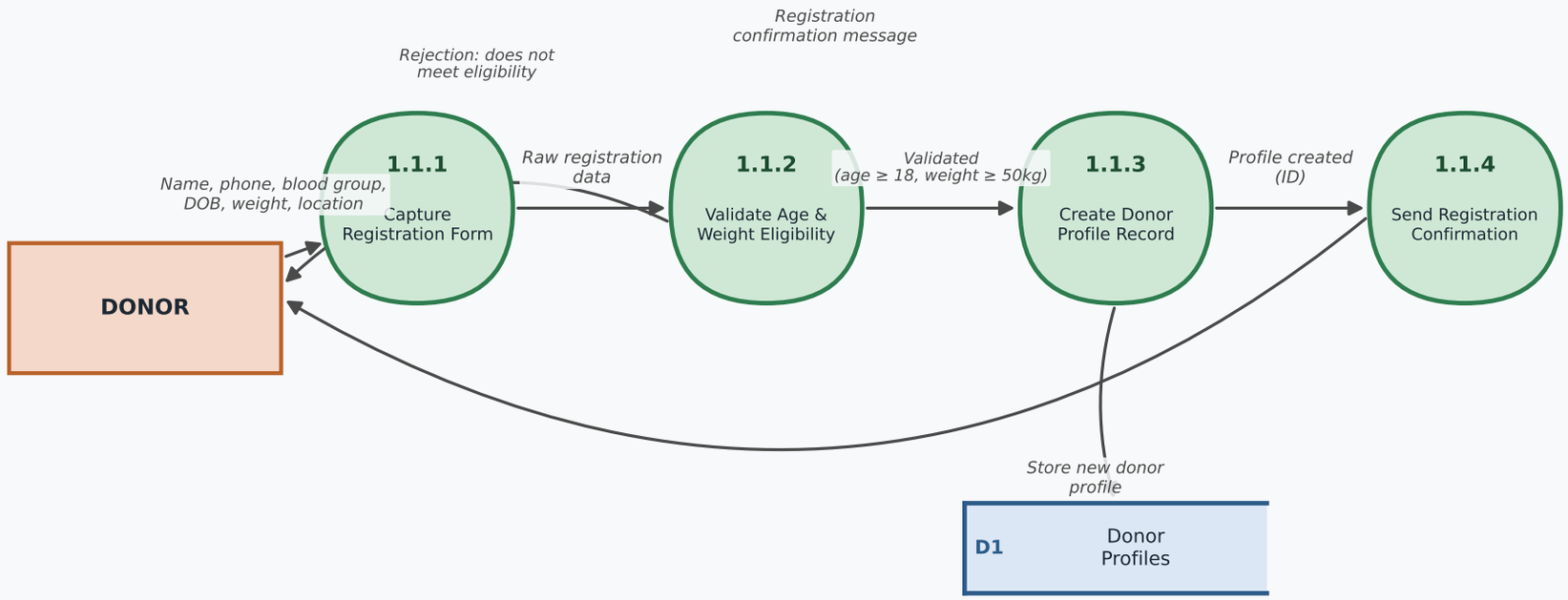
Level 1 DFD — decomposition of Process 6 (REQ-4.6.1 – 4.6.3)



Legend: rectangle = external entity | rounded shape = process | open box = data store | arrows = data flows

1.1 Register New Donor

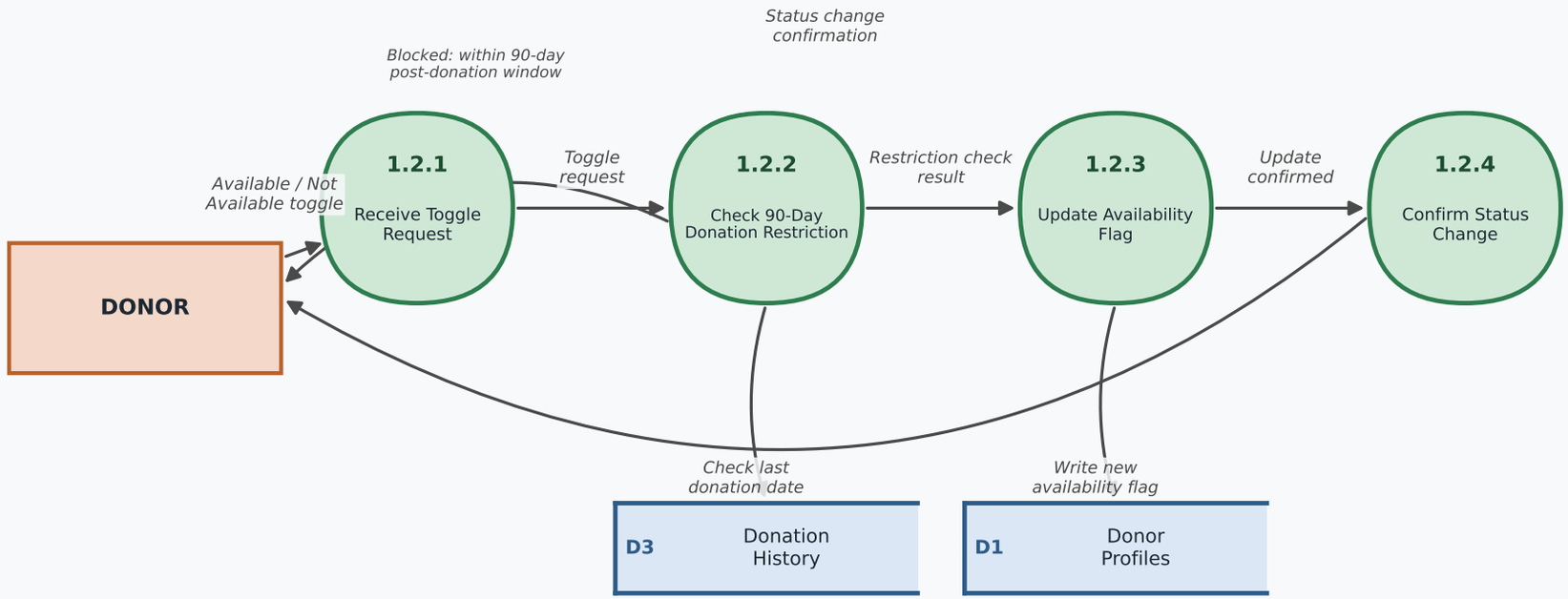
Level 2 DFD — decomposition of sub-process 1.1 (REQ-4.1.1, REQ-4.1.2)



Legend: rectangle = external entity | rounded shape = process | open box = data store | arrows = data flows

1.2 Toggle Availability Status

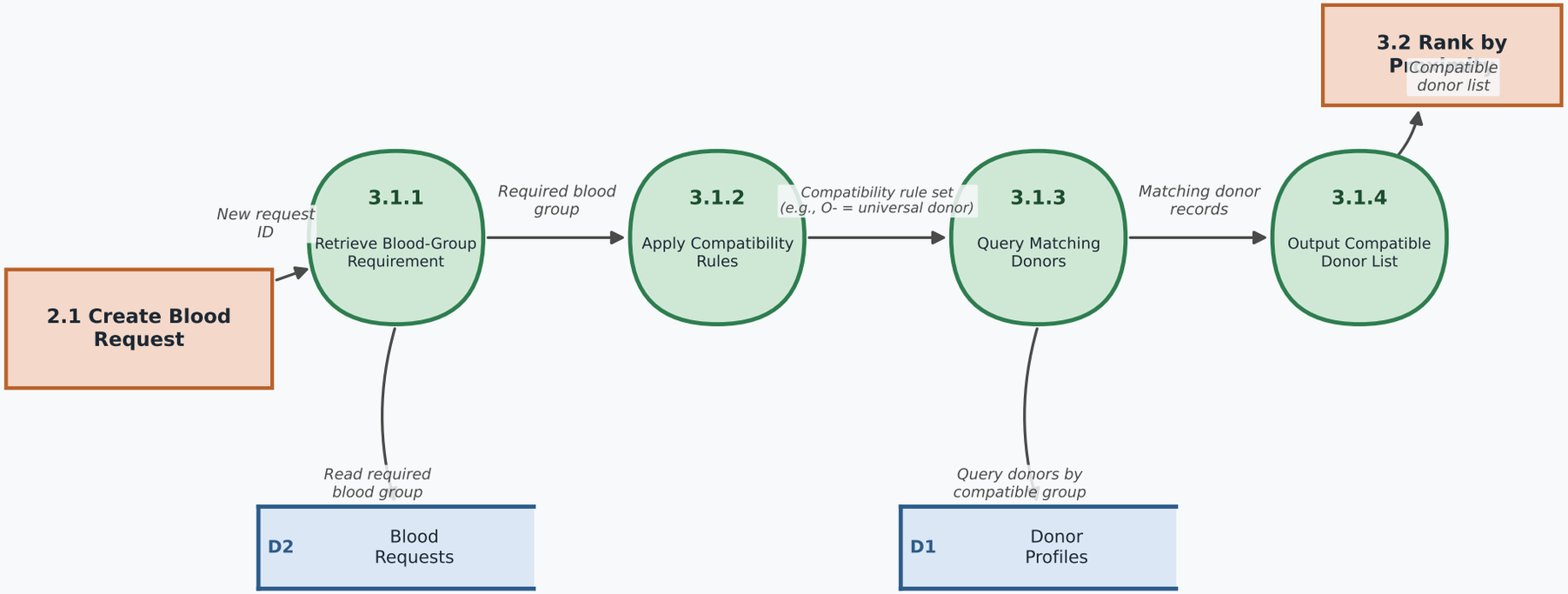
Level 2 DFD — decomposition of sub-process 1.2 (REQ-4.1.3, REQ-4.1.4)



Legend: rectangle = external entity | rounded shape = process | open box = data store | arrows = data flows

3.1 Filter by Blood-Group Compatibility

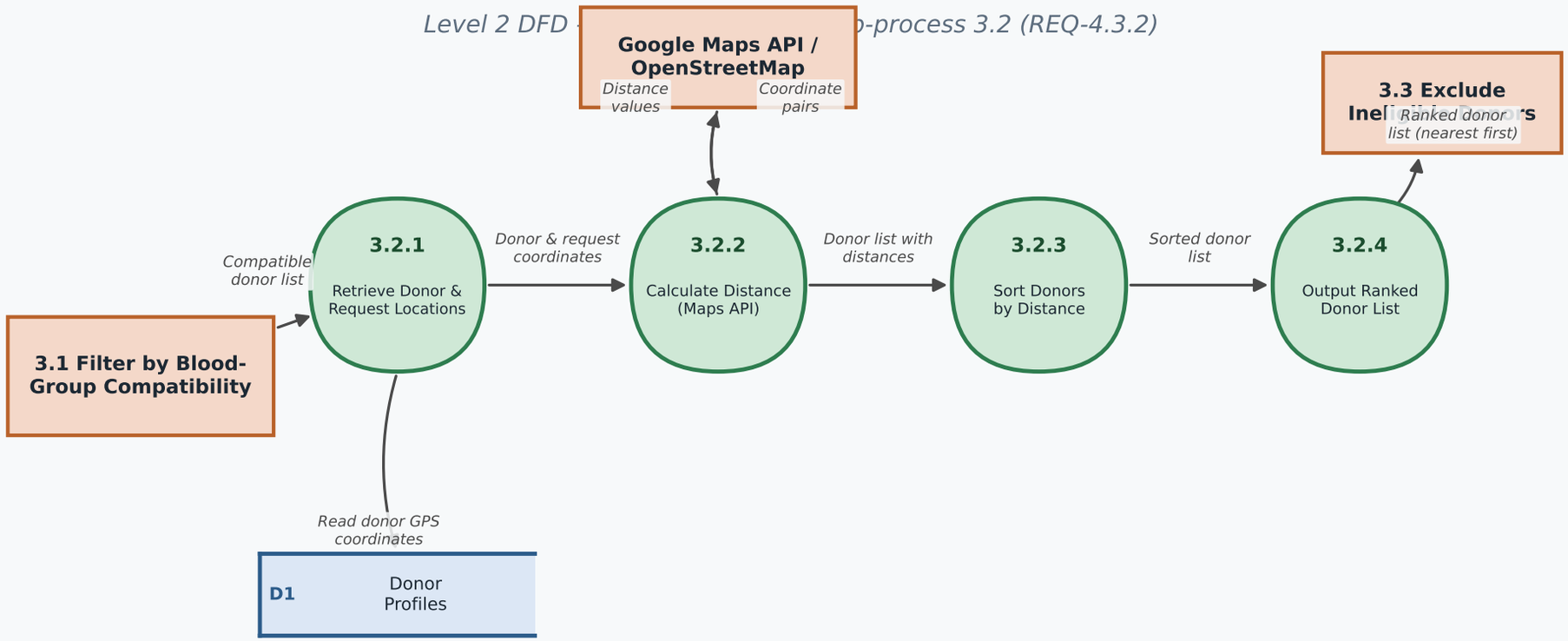
Level 2 DFD — decomposition of sub-process 3.1 (REQ-4.3.1)



Legend: rectangle = external entity | rounded shape = process | open box = data store | arrows = data flows | side boxes reference neighboring Level-1/Level-2 processes

3.2 Rank by Proximity

Level 2 DFD — sub-process 3.2 (REQ-4.3.2)



Legend: rectangle = external entity | rounded shape = process | open box = data store | arrows = data flows | side boxes reference neighboring Level-1/Level-2 processes & external API